

Common Ancestor

Binary Trees, Lowest Common Ancestor (LCA) · Mentor-Led Lab (Student Edition)

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Difficulty ★★★ Advanced

Problem

Given a binary tree and two node values, find their lowest common ancestor - the deepest node that has both values somewhere in its subtrees - and report the depth of that ancestor from the root.

Implement the LCA recursion and the depth lookup from scratch.

What to Compute

- LCA: the value of the lowest common ancestor of the two given nodes.
- Depth: the depth of the LCA from the root, measured in edges (the root is depth 0).

Input / Output Format

Input: Line 1: the tree in level order (values and N). Line 2: two integers A and B.

Output: Two lines: LCA and Depth.

Examples

Example 1

Input: tree = 3 5 1 6 2 0 8 N N 7 4, A = 7, B = 4

Output:

LCA: 2
Depth: 2

Explanation: 7 and 4 are the two children of node 2, so their lowest common ancestor is 2, which sits two edges below the root (3 -> 5 -> 2).

Example 2

Input: tree = 3 5 1 6 2 0 8 N N 7 4, A = 5, B = 1

Output:

LCA: 3
Depth: 0

Explanation: 5 and 1 are the root's two children, so their lowest common ancestor is the root 3 at depth 0.

Constraints

- Both A and B are present in the tree.
- The LCA is the deepest node with both A and B in its subtrees.
- Depth is measured in edges from the root (root = 0).

Sample Run

Sample Input

```
3 5 1 6 2 0 8 N N 7 4
7 4
```

Sample Output

```
LCA: 2
Depth: 2
```